

Hello and welcome to your home science experiment!

You are part of an important project in the Stenglein Lab to optimize fly traps for stability during shipping and fly collection. The data collected from this pilot program will inform how we conduct a broad public engagement experiment over the next five years. We hope that you will want to continue to be part of this project over the years to provide valuable data over time and location.

We are so excited that you are able to participate in our pilot study. We hope you enjoy the process and have fun with it. If you have any questions throughout the experiment please contact Lexi (Lexi.Keene@colostate.edu) or Tillie (Tillie.Dunham@colostate.edu).

There is an informational sheet discussing the basics of fruit fly life cycle and morphology, an expansion on our goals of this study, what we are going to do with the flies and how the data from this experiment will inform future studies included in the kit. Additionally, there is a section at the end of the instructions for you to give us feedback. We want to know what you think works and what doesn't and also any ideas you may have for experiments you would like to do at home that our kit could help fulfill. Please complete this as this feedback will aid us in improving the kit for next summer and beyond!

Package Contents

- 2 Jars with fly food
- 2 3D printed trap lids
- 2 Pieces of parafilm
- 1 Bag of yeast
- 1 return shipping label
- Instructions/feedback sheet
- Information sheet
- 1 Banana (to be purchased or used from personal stock)

Instructions for Setting up Fly Traps

1. We ask that you use a banana from your home or purchase one to complete the following steps.
2. Mash up one banana, add yeast and mix until it is a paste like mixture.
3. Remove the white lid and spread the banana yeast paste onto the surface of the food. (use a spoon or whatever you have around the house to spread the paste).
***Be sure to save the original white lids!**
4. Place the 3D printed trap lid or what we like to call the flyght house onto the jar.

Example:



5. Label the jars with sharpie. One should be labeled “*inside*” the other “*outside*”.
7. Place the jar labeled “*inside*” somewhere in your house. Possibly in your kitchen (around the trash can or fruit bowl) or near house plants, wherever you think will have more fruit flies.
***If you feel uncomfortable with catching flies inside your house, you may place both outside but make sure they are in different places. Maybe one is in the compost while the other is on your porch or patio. If you plan to do this, please write a clear description of the type of location the jar was in either on a separate piece of paper or on the jar itself.**
8. Place the jar labeled “*outside*” somewhere outside of your house where it won’t be disturbed. If you have a compost pile that will be an excellent choice. Around plants/trees will work well too.
9. Keep an eye on These traps for about 2 weeks. Below is an image of the type of fruit fly we are looking for. Other insects may get in the traps. That is okay, we will identify everything that gets to us.



https://commons.wikimedia.org/wiki/File:Drosophila_melanogaster_Proboscis.jpg

10. After the 2 weeks (you may leave it longer if the food seems to be holding up or not as long if you catch a bunch of flies!) follow the instructions in the next section

***If you're not sure if the trap is ready, email us a picture and we can help you out!**

Please view the following video if you have never used parafilm before:

https://www.youtube.com/watch?v=sDKPqHgF_Gw

Instructions to Finish Collecting

1. Once flies have laid eggs in the jar and you are ready to end the experiment you will need to switch the caps as you did at the very beginning. We recommend doing this one jar at a time.
2. Take the jars from “*inside*” and “*outside*” locations and place one of them in your freezer for **1-1.5 minutes** (no longer or the flies will die). During this time get the original lids ready for the switch. You will need to be quick. This causes the flies to go to sleep but not for long!

3. Open the freezer, gently tap the jar on a counter so the flies fall to the bottom and quickly switch the lid from the trap lid back to the original white lid. If flies escape that is OK - you may want to do this outside in case any do escape.
4. Repeat steps 2 and 3 for the second jar.
5. Repackage the jars using the same box that was sent to you, there should be a return label in the package that you can attach. Then ship it out to us here in Fort Collins.

Thank you so much for participating! We can't wait to see the flies and hear the feedback from you!

Feedback Section:

1. How clear are the instructions? Are there steps that are unclear or don't make sense? Is the wording appropriate?
2. Is there a step you feel is missing?
3. Any other comments on the instructions? What would you like to see included, what could be removed, etc...

4. Was the information on the information sheet sufficient?
5. What other questions do you have about the experiment?
6. Was the wording on the informational sheet appropriate?
7. Is there something else you want to see on the information sheet?
8. Any other comments, questions, or concerns you would like us to know about?

Background Information- Why do we care and why should you?

Why fruit flies?

Drosophila melanogaster (*D. melanogaster*) fruit flies have been the focus of biologists since the early 1900s. These flies are ideal for use in the lab due to their small size, quick life cycle, simple reproduction and low maintenance costs. Not to mention, they share roughly 70% of genes involved in human disease allowing researchers to study human disease in a practical and safe manner.



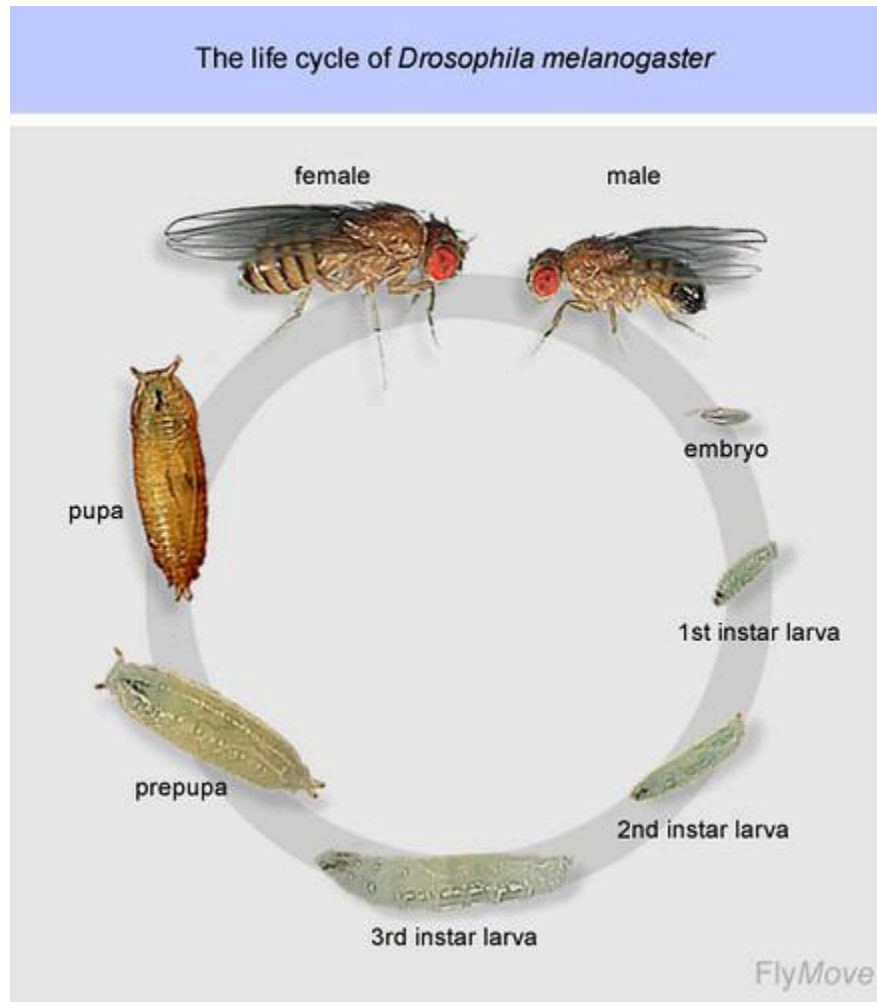
Thomas Hunt Morgan's Fly Room in the 1920s, 613 Schermerhorn Hall
https://en.wikipedia.org/wiki/Schermerhorn_Hall#/media/File:Schermerhorn613.jpg

The Stenglein lab at Colorado State University uses *D. melanogaster* to study viruses. Viruses infect all living organisms, including tiny insects like fruit flies, and, as strange as it may seem, important fundamental discoveries about the immune system have been made studying viruses that infect flies!

In addition to being well suited for laboratory-based studies, fruit flies are common and easy to catch in nature. For this project, we will use volunteer scientists like yourselves to catch fruit flies in locations around the United States. This will enable

us to study the viruses that naturally infect fruit flies and learn even more about how viruses and their hosts interact.

Some information on D. melanogaster:



The Life Cycle of *Drosophila melanogaster*
<https://www.flickr.com/photos/11304375@N07/2993342324>

Above is a cartoon of the *D. melanogaster* life cycle. Its short simple life cycle makes it easy to rear in the lab. The flies you catch will hopefully lay eggs before you ship them back so that we will have a new generation of live flies when they arrive at our lab. When you switch caps when getting ready to send them back, take a look in the jar and see if you can see eggs or larvae, which look like small worms!

In the image above you can also see information on how to determine the sex of the flies you caught. Generally females are larger and have a more oval shaped abdomen.

Males also have more black on their abdomen. Can you tell whether you've trapped males or females? A magnifying glass or using zoom on your phone's camera may help!

What is in the food?

The food in the jars is safe, attractive to flies, and has been optimized to be stable during transport and collection. The main components of the food includes cornmeal, agar, water and various sugars like corn syrup. Additionally, an antifungal agent is included to increase the lifespan of the food especially in hot and humid environments.

We also have included an additive shown to attract more flies! We want to collect as many flies as possible and spent the better part of this summer testing various additives. The winner of these experiments was banana extract (banana flavoring you can buy at the grocery store) and we've added some of it to the food. If you smell the food you will get whiffs of cornmeal and the antifungal agent but maybe you will smell the banana too!

Could I make a fly trap at home?

Yes! Flies love bananas and yeast, they truly can't get enough of it. If you want to continue asking questions and studying the humble fruit fly, all you need is a container you can poke some holes in, a banana and yeast. You can leave it outside or in your home and quickly capture all sorts of flies.

If you want to catch your own flies at home but want to use one of our jars and caps, please email Lexi or Tillie so we can get that to you!

If you want to challenge yourself, try to identify all the flies you catch! You can use this guide to help you: https://biologicalsurvey.ca/ejournal/mmg_31/mmg_31.pdf

Why do I need to put them in the freezer?

Placing the flies in the freezer puts them to sleep for a little bit. We do this because taking the cap off when they are still awake will result in the escape of some flies. The important thing here is that we don't want to kill the flies which may happen if the flies are left in the freezer for too long so be careful not to leave them for longer than 2 minutes.

What are you doing with them?

We are going to use these flies to study viruses that naturally infect fruit flies. Surprisingly, even healthy flies are infected by several viruses most of the time (don't worry, none of these infect humans). The most prominent virus we will be looking for is called galbut virus, which infects more than 50% of fruit flies globally. To do this we will extract RNA from each fly and screen it to see if galbut virus is present or not. Some other viruses we will be looking for include the Vera, La Jolla, Thika and Nora viruses.

Why do you care about these viruses?

We are mostly concerned with the presence of galbut virus because it is so good at infecting flies. This has piqued the interest of the lab for many reasons. An important finding about this virus is that it is transmitted vertically, that is from parent to offspring, with near 100% efficiency.

What will the data from this experiment be used for?

The data from the experiment you are involved in this summer will help to inform us of the level of galbut virus of fruit flies across the United States. Galbut virus is a Partitivirus and is highly prevalent across many populations in the U.S. The goal of this outreach project is to collect flies in order to test for Galbut virus. We are also trying to get citizens involved in scientific research and open up the scientific community to the general public.